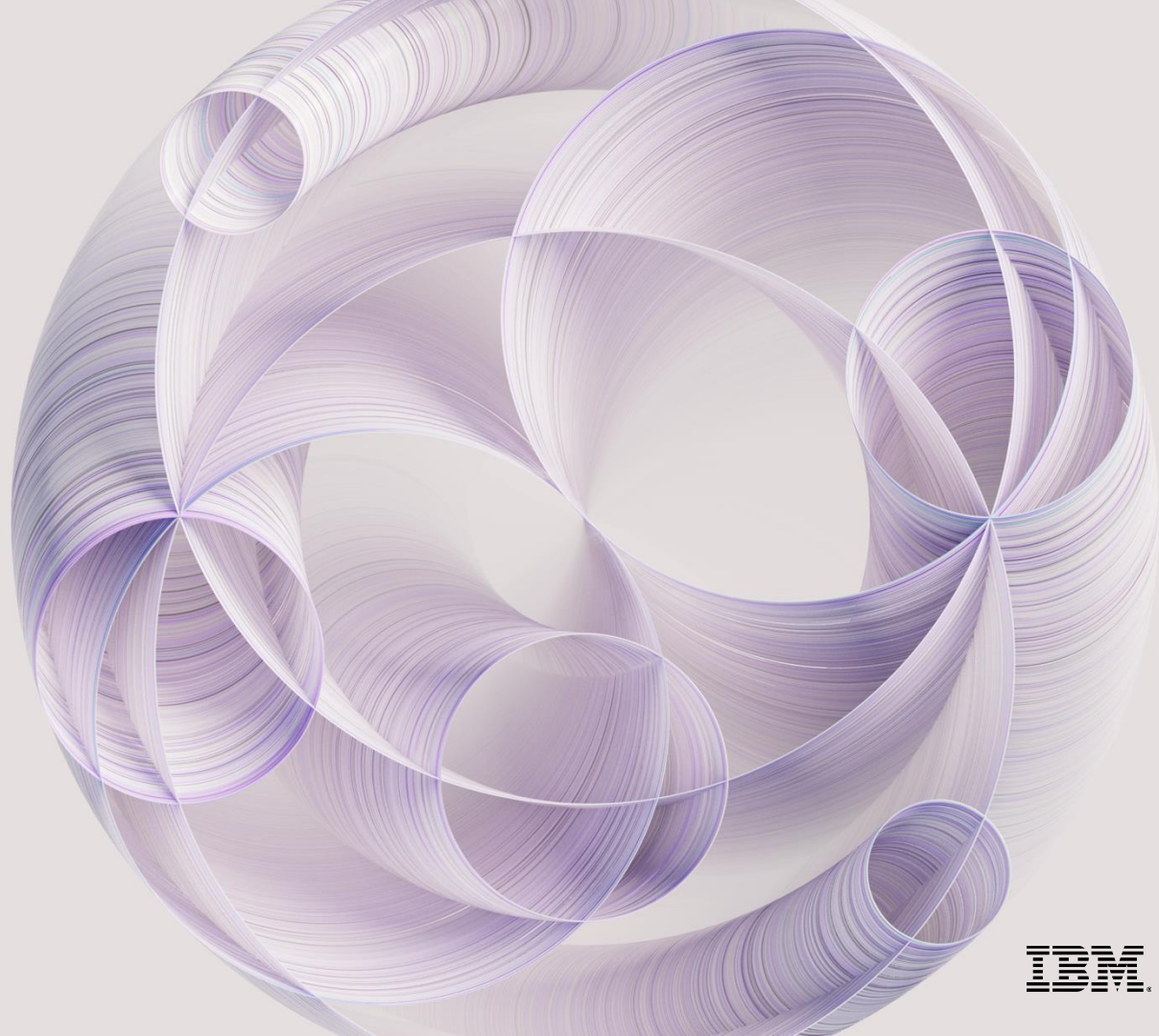


Streamline the
development of
AI applications
with watsonx.ai

watsonx.ai



IBM

IBM Software

Hybrid Cloud

Unify on-prem, public, private clouds and edge to scale virtualization and AI across environments



Transaction Processing

Deliver unmatched transactional performance, security and reliability



Automation

Automate technology lifecycle management with AI for productivity, resiliency and spend optimization



Data

Access trusted and secure data to drive AI productivity



AI

Open and Trusted



AI Productivity

Reinvent how work is done with
AI agents/assistants

- AI Assistants

AI/ML Ops

Work with AI models, tools and
governance that's built for
business—engineered to ensure
trust and scalability in
applications

- AI Models
- AI Tools
- AI Governance

Data Fabric

Bring all your business data
together and optimize how it
moves through your systems to
scale analytics and AI in your
applications while protecting it.

- Databases
- Data Intelligence
- Data Integration
- Data Security

Data Storage

Store data across edge,
core and clouds

- Software-defined Storage

Data

Access trusted and secure data to drive
AI productivity



AI Productivity

Reinvent how work
is done with AI
agents/assistants

AI Assistants



watsonx Code
Assistant™



watsonx
Orchestrate™



Planning
Analytics

AI/ML Ops

Work with AI models, tools and governance that's built
for business—engineered to ensure trust and scalability
in applications

AI Models



Granite™



Meta Llama



Mistral

AI Tools



watsonx.ai™

AI Governance



watsonx.
governance™

Data Fabric

Bring all your business data together and optimize how
it moves through your systems to scale analytics and AI
in your applications while protecting it

Databases



watsonx.data™

Data Intelligence



Data Product
Hub



Knowledge
Catalog



Manta Data
Lineage

Data Integration



DataStage®



Databand®



Streamsets

Data Security



Guardium® Data
Security Center

Data Storage

Store data across
edge, core and clouds

Software-defined Storage



Storage Ceph®

watsonx

A portfolio of AI products that accelerates the impact of generative AI in core workflows to drive productivity.

watsonx.ai

Enterprise-grade AI studio that helps AI builders innovate with all the APIs, tools, models, and runtimes to build AI solutions

Featuring **IBM Granite**, and popular third-party models including **Mixtral**, **Llama** series

watsonx.data

The **hybrid, open data lakehouse** to power AI and analytics with all your data, anywhere

watsonx.governance

End-to-end toolkit for AI governance to manage **risk and compliance** across the entire AI lifecycle.

watsonx Orchestrate

An enterprise-ready solution that helps create, deploy, and manage AI assistants and agents to automate processes and workflows.

watsonx Code Assistant

Accelerate development, **application modernization**, and assist with IT Operations

Why IBM watsonx for scaling enterprise AI to drive productivity

Open

- Offers choice to train the right foundation models, including open-source models, and the choice of data, tools, and frameworks to achieve desired business outcomes.
- Run AI wherever the business needs to, across any cloud, at scale.

Trusted

- Built with open and transparent technology to give enterprises confidence in their AI and meet regulatory compliance demands.
- Responsible AI and protected data backed by enterprise governance and security controls.

Integrated

- Integrates technology seamlessly into existing infrastructures, systems, and processes with choice of cloud to transform the enterprise and drive productivity from within.
- Embedded AI for targeted use cases that drives enterprise scale productivity.

Streamline the development of AI applications with watsonx.ai



Model selection



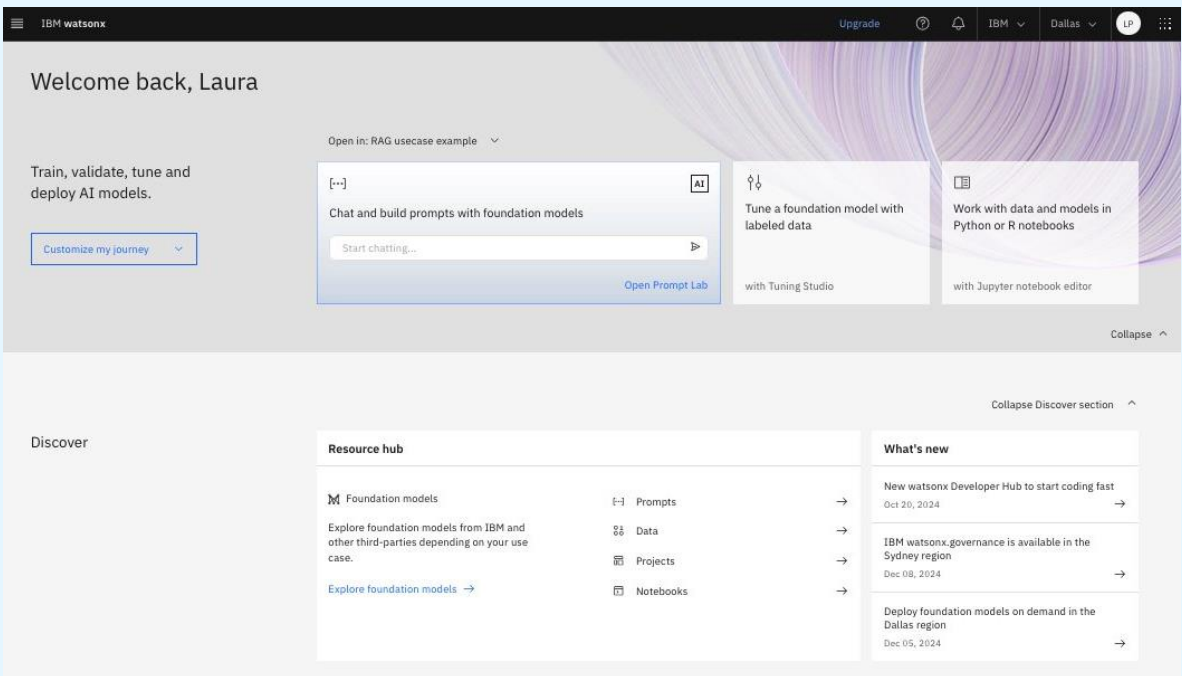
Model customization



AI development and deployment

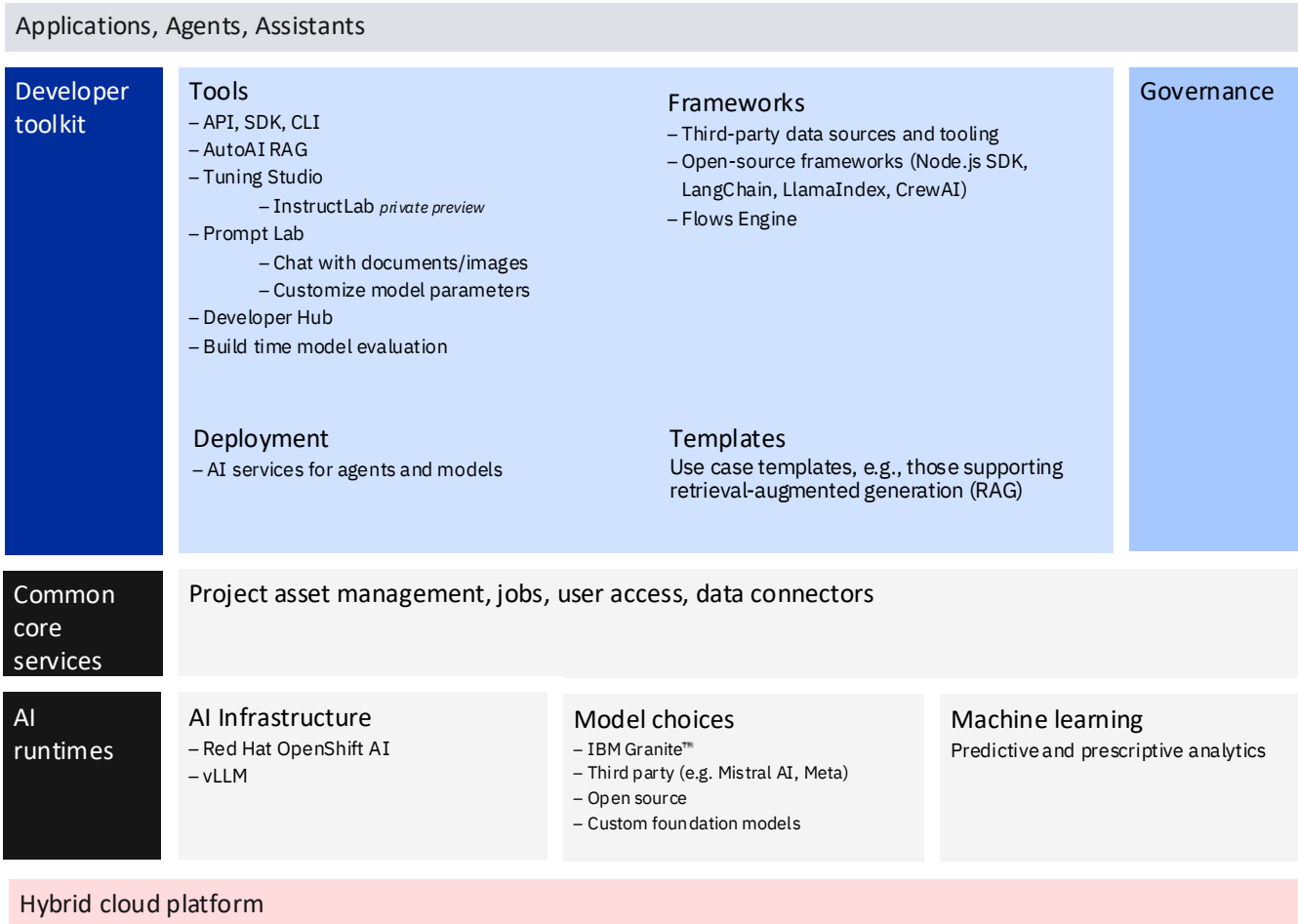


AI governance



The enterprise-grade
AI developer studio

Integrated AI platform
with all the APIs, tools,
models, and runtimes to
simplify and scale the
development and
deployment of AI

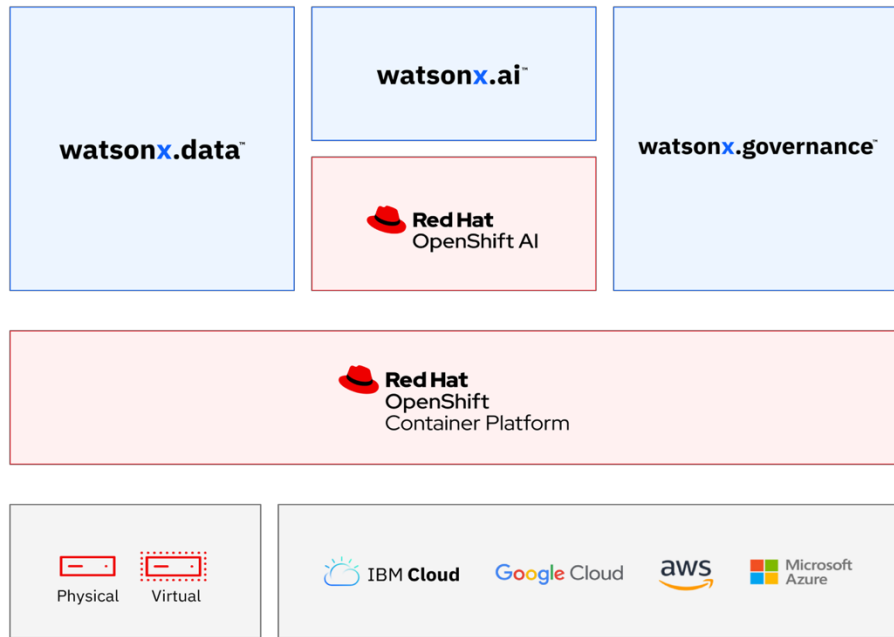


watsonx.ai and Red Hat

Watsonx.ai integrated with Red Hat OpenShift AI & Red Hat OpenShift Container Platform delivers an accelerated time to value through an optimized generative AI and ML workflow.

Key benefits:

- A complete platform for managing AI workloads, training models and scaling AI deployments
- Access to high quality, governed, and trusted AI through curated IBM-developed, third-party, and open-source foundation models
- Deploy to on-premises environments, private or public clouds
- Integrations to the wider watsonx and Cloud Pak for Data platforms.



Creating Generative AI Solutions

Model Selection

watsonx.ai Provided Models

A highly curated library with IBM-built, open-source and third-party foundation models that are pre-deployed and ready to use in watsonx.ai for a wide range of business requirements and budgetary considerations. These models are deployed on multi-tenant GPU's.

Key benefits:

- Great for experimentation and prototyping to determine which model is best for your use case
- Pay only for the tokens that are consumed, no time commitment
- Ideal for clients that need periodic or infrequent use of foundation models
- Indemnification offered for IBM and select models

Learn more in [documentation](#)

Most models available in all regions

Select a foundation model

Select a model that best fits your needs. All provided models support English text. Check the model information for other supported languages.

Search for a model or task

Model Name	Provider	Type
granite-13b-chat-v2	IBM	Provided
granite-3-1-2b-instruct	IBM	Provided
granite-3-8b-instruct	IBM	Provided
granite-3-2b-instruct	IBM	Provided
granite-3-1-8b-instruct	IBM	Provided
llama-3-3-70b-instruct	Meta	Provided
granite-guardian-3-8b	IBM	Provided
mistral-large	Mistral AI	Provided
granite-guardian-3-2b	IBM	Provided
pixtral-12b	Mistral AI	Provided
granite-20b-multilingual	IBM	Provided
codestral-22b	Mistral AI	Provided

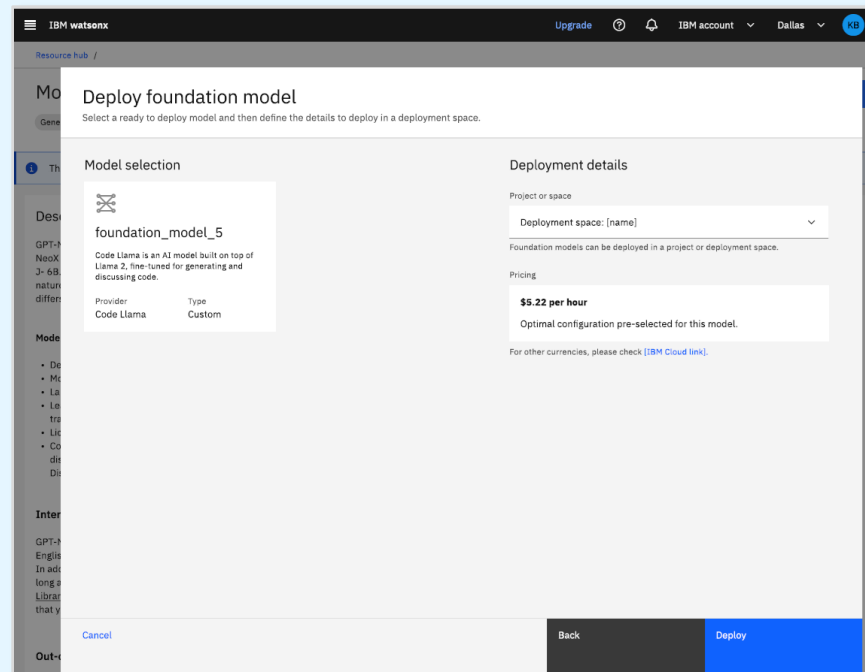
Deploy on Demand Models

A curated collection of high quality and popular foundation models that can be easily deployed to a dedicated GPU with just a few clicks.

Key benefits:

- Faster interactions with a dedicated, single-tenant deployment, hosted 24/7 until de-provisioned by client
- Predictable hourly hosting price compared to variable token-based pricing
- Supports full context length of the model
- No rate limits on the inference requests per second (vs. 8 req/s limit on pre-deployed models)

Learn more in [documentation](#)



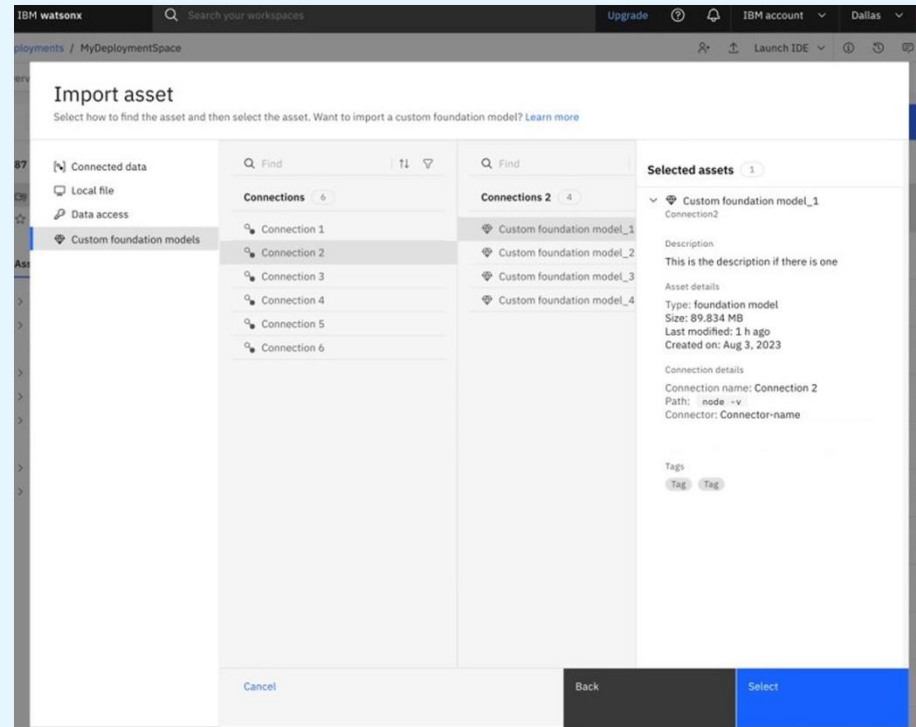
Custom Foundation Models

Import custom foundation models for maximum flexibility in how generative AI solutions are developed.

Key benefits:

- Leverage externally fine-tuned LLMs that support a specific language or are customized for an industry or business domain
- Import a model that is not already provided by IBM.
- Compatible with 1000's of models from repositories like Hugging Face, which provides access to a huge selection of open-source foundation models

Learn more in the [blog](#), [tutorial](#) or [documentation](#)



Access **1000+ models** in watsonx.ai from a mix of platform-provided models and custom foundation models available for import

	Provided Models (Ready to use models)	Deploy on Demand	Import Custom Foundation Models	Software
Overview	Multi-tenant SaaS offering where capacity is shared	Single-tenant SaaS offering with curated set of models from IBM	Single-tenant SaaS offering with customer-provided models	On-premises deployment on customer infrastructure
Cost Structure	Token-based pricing, billed on both input and output tokens; available for all SaaS Plans	Hourly pricing - includes hosting and inference; billed while model is deployed; Standard Plan only	Hourly pricing - includes hosting and inference; billed while model is deployed; Standard Plan only	No additional model pricing; client is billed on software entitlement
Benefits	- Pay only for the tokens that are consumed - Indemnity offered for IBM and select models	- Predictable cost based on hourly model deployment - Dedicated GPU capacity - Ease of deployment - Indemnity offered for IBM and select models	- Predictable cost based on hourly model deployment - Dedicated GPU capacity	- Maximum flexibility in choosing which models to deploy and on what kind of GPUs - Air-gapped environment for secure workloads
Drawbacks	GPU capacity is shared; unable to take full advantage of the throughput	Deprovision the model when not in use to optimize costs	- Requires user to import relevant model files - Deprovision the model when not in use to optimize costs	Infrastructure setup required
Best Fit for	- Experimentation - Small companies with infrequent use	- Large production workloads - Enterprises with SLA requirements	- Large production workloads - Enterprises with SLA requirements	- Large production workloads - Enterprises with SLA and security requirements

Model Customization

watsonx.ai Tuning Studio

Improve foundation model performance through customization using a variety of methods:

- Fine Tuning
- Prompt Tuning
- InstructLab

Easy to use, step by step interface to accelerate the setup time.

Customized models can be deployed within watsonx.ai and inferred through the Prompt Lab or API.

Learn more in [documentation](#)

The screenshot displays the IBM watsonx.ai Tuning Studio interface for creating a new tuned model. The top navigation bar includes the IBM watsonx logo, a search bar, and user account information. The main content area is titled 'New tuned model' and is divided into two columns: 'Configure details' and 'Submission details'.

Configure details:

- Which foundation model do you want to tune?** A dropdown menu shows 'granite-13b-instruct-v2'.
- Which customization method do you want to use?** Three options are available: 'Prompt tuning' (adjusts input prompt), 'Fine tuning' (adjusts model parameter weights), and 'InstructLab' (adjusts model's taxonomy). 'InstructLab' is selected.
- How do you want to work?** Two buttons are shown: 'No code' (selected) and 'Code'.
- Add a skill or knowledge to the model's taxonomy** A button labeled 'Add skill or knowledge' with a plus icon is present.

Submission details:

- Submit for training** A section with instructions for reviewers to check grounding data. It includes a 'GitHub ID' field with the value 'jane.doe' and a link to 'Learn more'.
- New skill: Arithmetic_w_grammar** Two entries are listed, each with a description: 'To teach a large language model about grammar and arithmetic'.

A blue 'Submit for review' button is located at the bottom right of the interface.

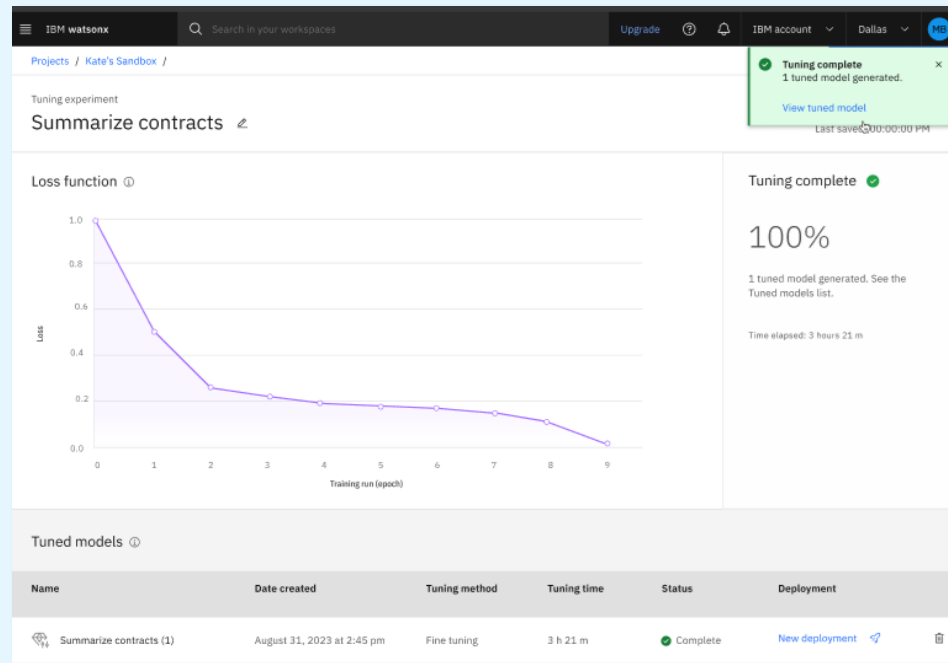
Tuning Studio: Fine Tuning

Unlock the full potential of foundation models by customizing it for a specific task, use case or business domain with structured data.

Key benefits:

- Improved performance and accuracy
- Reduced Bias from original training data
- Easy to use interface that supports structured training data in files or connected data sources
- Adjust parameters and view performance metrics

Learn more in [documentation](#)



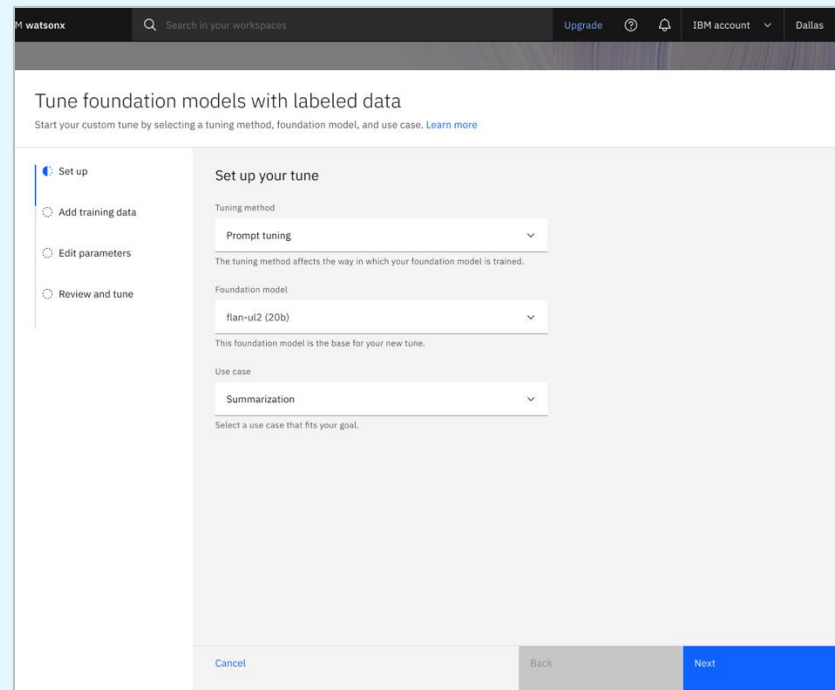
Tuning Studio: Prompt Tuning

Leverage Prompt Tuning to further train a model with focused, business data.

Key benefits:

- Efficient, low-cost way of adapting an AI foundation model to new tasks
- Tune the model without changing the underlying base model or weights
- Achieve comparable results to full fine-tuning depending on base model size and data used
- Minimize the risk of catastrophic forgetting

This tuning method is a subset of Parameter Efficient Fine Tuning (PEFT). Learn more in [documentation](#).



The screenshot displays the Watsonx Tuning Studio web interface. At the top, a dark navigation bar includes the 'watsonx' logo, a search bar, and links for 'Upgrade', 'IBM account', and 'Dallas'. The main heading is 'Tune foundation models with labeled data', followed by a subtext: 'Start your custom tune by selecting a tuning method, foundation model, and use case. [Learn more](#)'. A left sidebar contains a 'Set up' section with three steps: 'Add training data', 'Edit parameters', and 'Review and tune'. The main content area, titled 'Set up your tune', features three dropdown menus: 'Tuning method' (set to 'Prompt tuning'), 'Foundation model' (set to 'flan-ul2 (20b)'), and 'Use case' (set to 'Summarization'). Below the 'Use case' dropdown is the instruction 'Select a use case that fits your goal.' At the bottom of the form are three buttons: 'Cancel', 'Back', and 'Next'.

Tuning Studio: Prompt Tuning

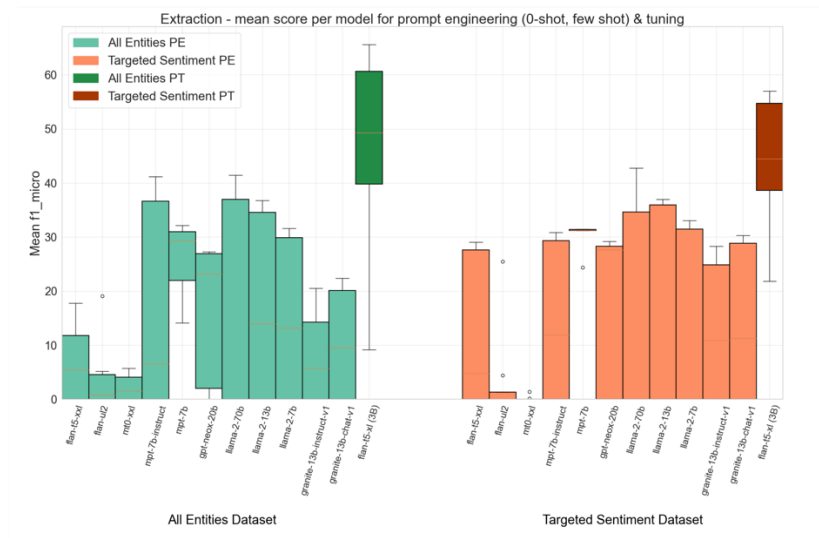
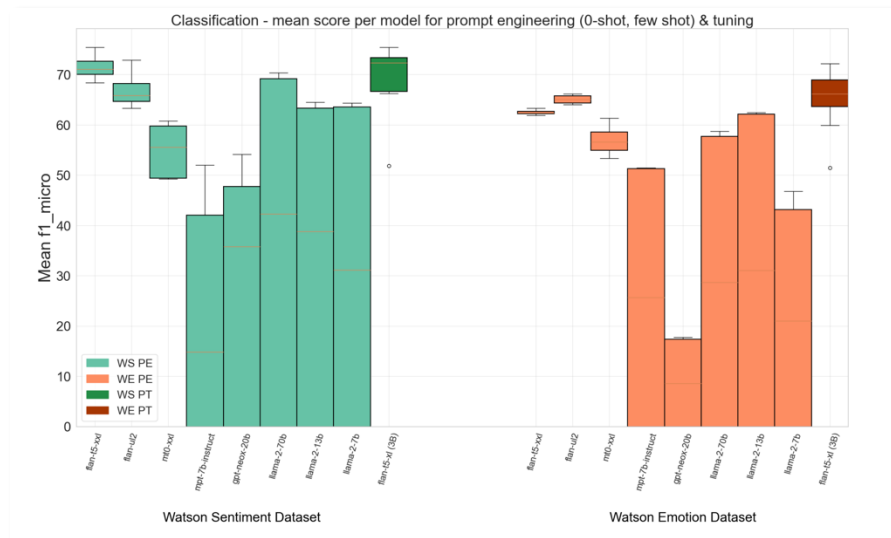
Why choose Prompt-tuning?

Performance lift:

- Tuned models perform on-par with much larger LLMs on NLP tasks like classification, entity extraction, summarization, etc.
- Prompt-tuning is also comparable to full fine-tuning in certain tasks/scenarios ([see whitepaper](#))

Cost and compute efficiency:

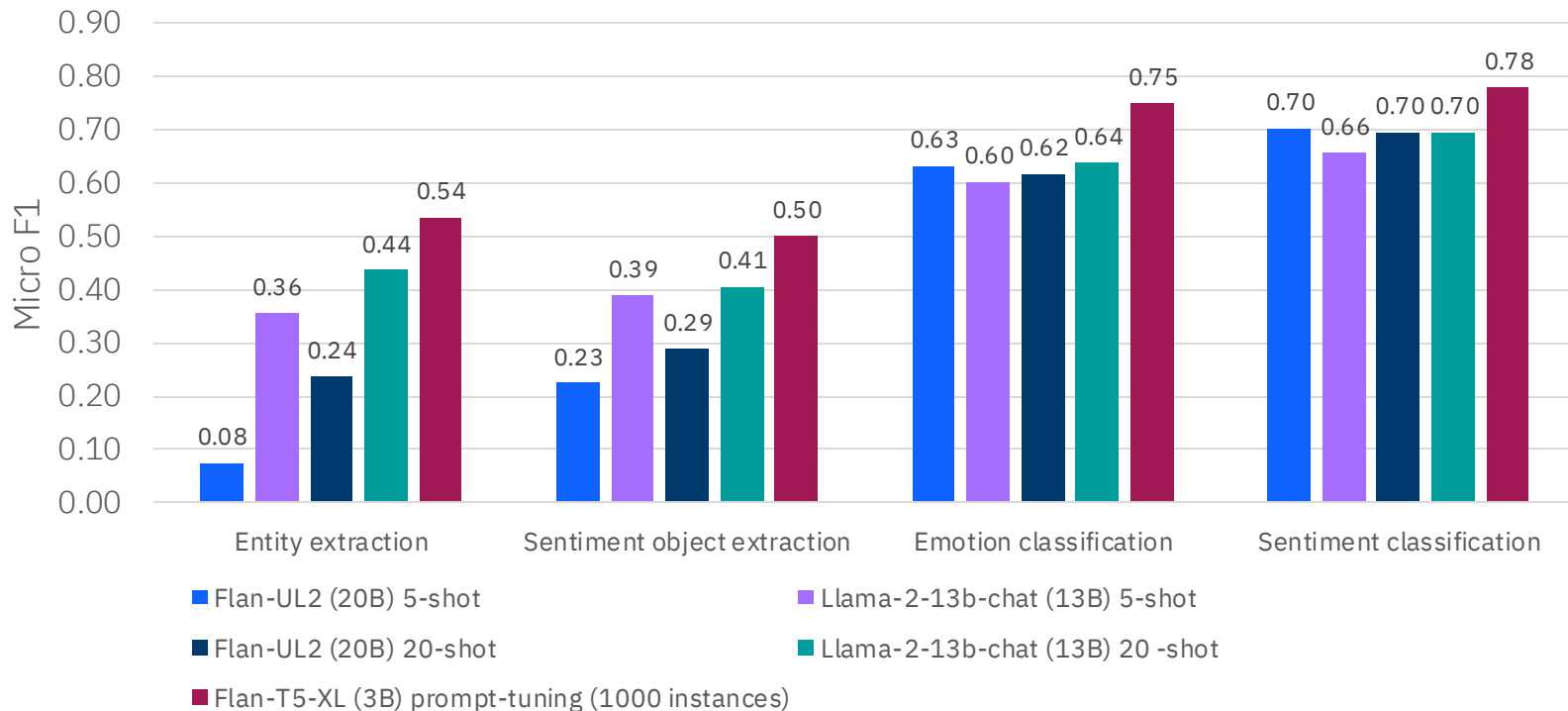
- Weights and parameters of underlying LLM are not adjusted during prompt-tuning
- Therefore prompt-tuned model does not need dedicated GPU for serving (instead just the soft-prompt vector output)
- In addition to smaller resource footprint post-tuning, training task itself is much less GPU intensive than fine-tuning



Tuning Studio: Prompt Tuning

Why choose Prompt-tuning?

Prompt engineering vs. Prompt-tuning in watsonx.ai



Tuning Studio: InstructLab

End to end private skills/knowledge submission, multi phase fine tuning, high fidelity synthetic data generation

- Ingest grounding data through documents in various formats (leveraging Watson Document Understanding)
- Submit skills/knowledge to a private taxonomy through easy-to-use interfaces or command line interfaces
- Conduct a peer review process on Github
- Multi-phase aligned fine tuning of models using synthetic data (knowledge, foundational skills, compositional skills)



Advanced
data ingestion



Data curation
and lineage



Data and
model evaluation



Developer
AI toolkit



Audit
readiness



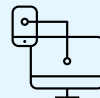
Collaborative
contribution



Rapid
iteration



Cost
savings



Unified
experience



Out-of-the-box
integrations

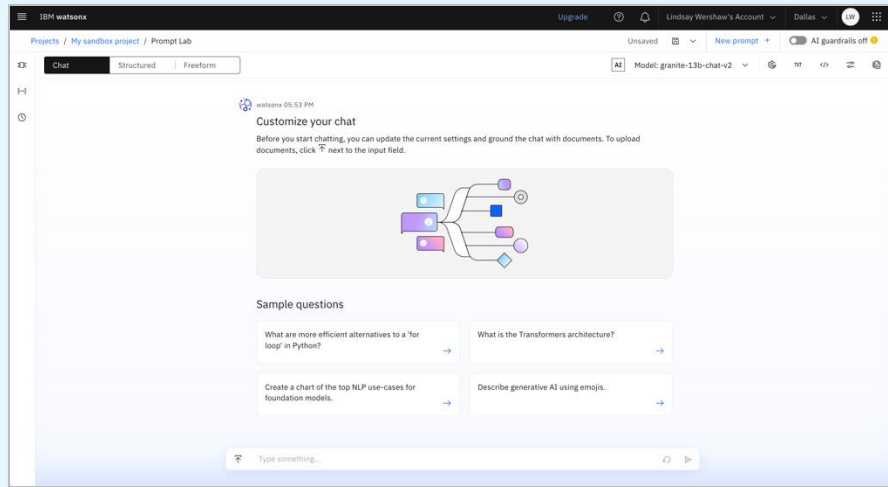
Prompt Lab

watsonx.ai Prompt Lab

Experiment with foundation models through an interactive user interface or an API.

- Easy to use chat, structured and free form prompt building interfaces
- Experiment with zero-shot, one-shot, or few-shot prompting to get the best results
- Adjustable AI guardrails
- Includes prompt examples for various use cases and tasks
- Save and share prompts for team collaboration
- Wide selection of foundation models to meet any task requirement
- Adjust model parameters to optimize results such as sampling, min/max tokens, stop sequences, repetition penalties, and more
- Export prompts and settings to a notebook to jump start development

Learn more in [documentation](#)



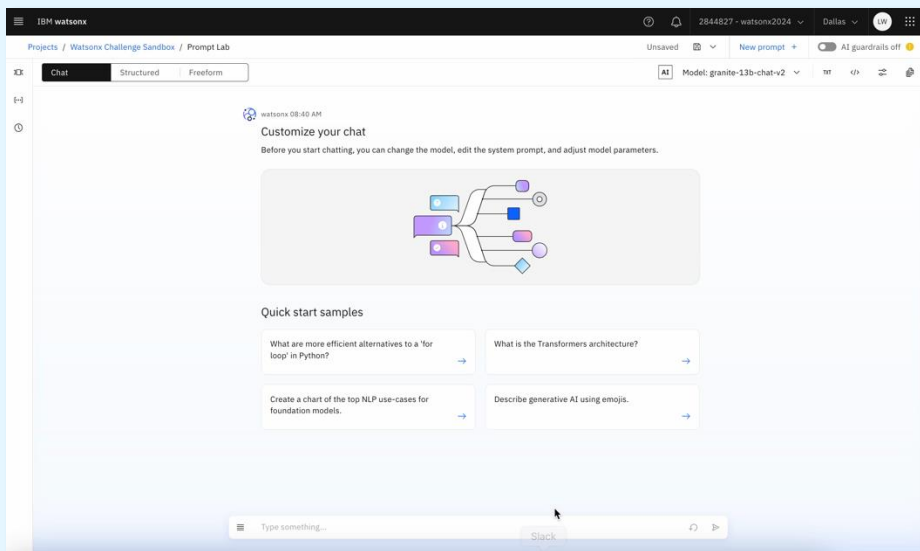
Chat with Documents

The Chat with Documents feature, within the Prompt Lab, provides an easy way to support RAG use cases.

Key benefits:

- Dramatically improve model output by grounding with relevant content
- Fast prototyping of your RAG use cases (e.g. test different LLMs, iterate on document index/collections, change embedding models, set chunk size and overlap, and more)
- Accelerate development by exporting the prompt as a python function deployed as a REST API
- Save prompts and settings as a template for team collaboration.

Learn more in [documentation](#)



Chat with Images

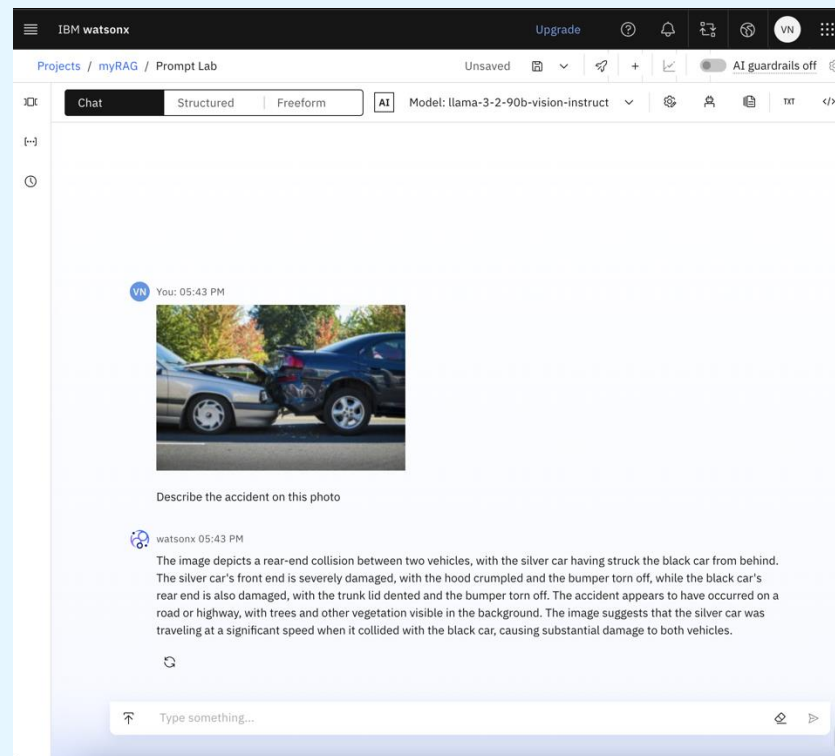
The Chat with Images feature, within the Prompt Lab, provides an easy way to convert visual information into text.

Example use cases:

- Automate the generation of alternative text for images to meet accessibility requirements
- Summarize photos to support use cases such as insurance claims (and many others)
- Convert images from a document into text before the document is used as grounding information for a RAG use case.

Chat with Images is supported by **Llama 3.2** (11b and 90b vision models).

Learn more in the [documentation](#) and [release blog](#).



RAG Solutions

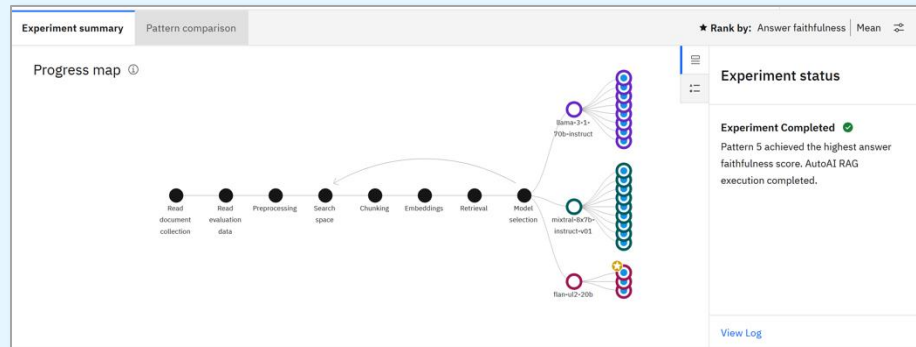
watsonx.ai AutoAI RAG

RAG prototypes are easy to build but hard to productionalize, which can require a team of experts and months of effort. AutoAI RAG helps to resolve this.

Key benefits:

- Accelerates design and deployment of optimized RAG systems based on client data and use case
- Quickly run various experiments to evaluate a constrained set of configuration options (models, chunk size, and more)
- Re-evaluate and modify recommended configurations when something changes (e.g. a new model version is released, quality of model responses change).

Learn more in [documentation](#).



AutoAI RAG tests a range of parameters (models, chunk size, etc.) in a series of experiments to automatically find the most optimal combination. It has 3 layers:

- Efficient RAG **hyper-parameter optimization** algorithm with end-to-end automation
- Best-of-breed RAG **evaluation metrics** and **benchmarking tools**
- **Parameterized RAG pipelines** for creating embeddings and for retrieval-based inference

watsonx.ai Text Extraction API

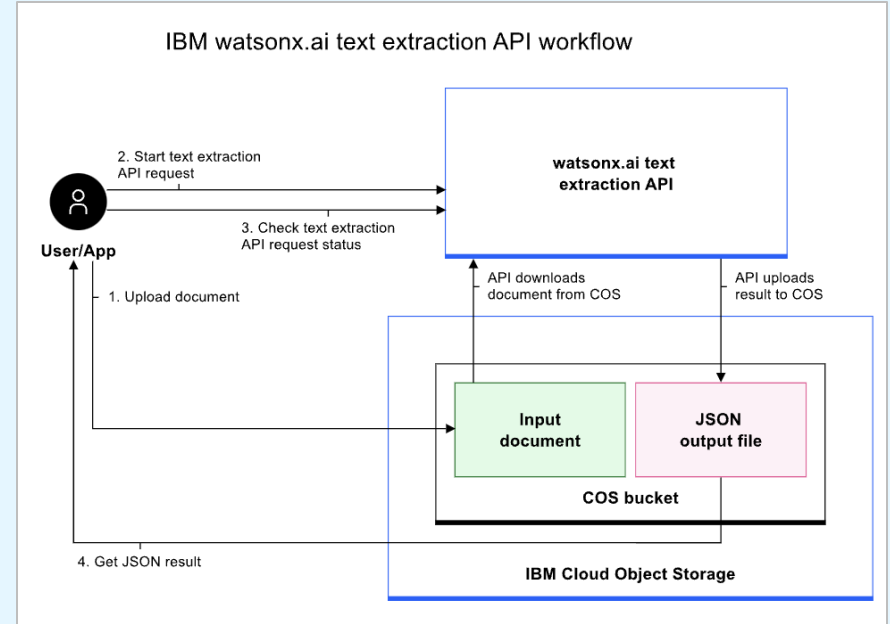
Extract content from documents which is essential for RAG use cases.

- Convert files with tables, diagrams and images into an AI-model friendly JSON file format
- Process the following file types: GIF, JPG, PDF, PNG, TIFF. Including scanned, hand-written documents
- Supports multiple languages

Technology:

- IBM's Natural Language Understanding (NLU) Service. See [Watson Document Understanding](#).
- Optical Character Recognition (OCR) to extract text from images.

Learn more in the [documentation](#).



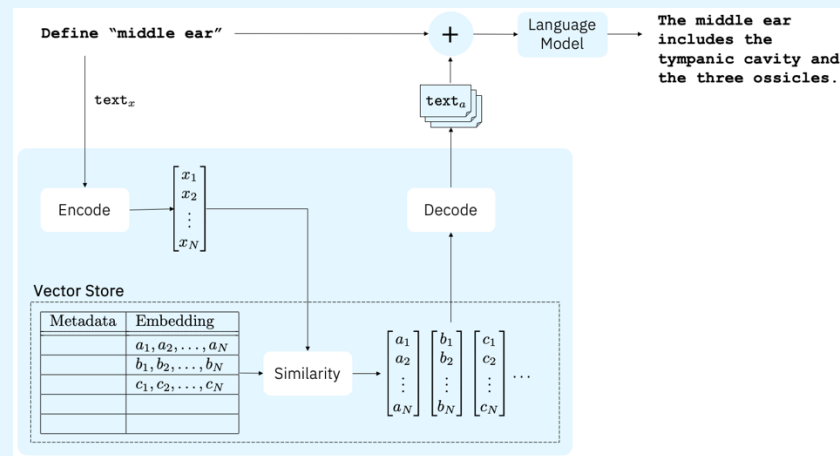
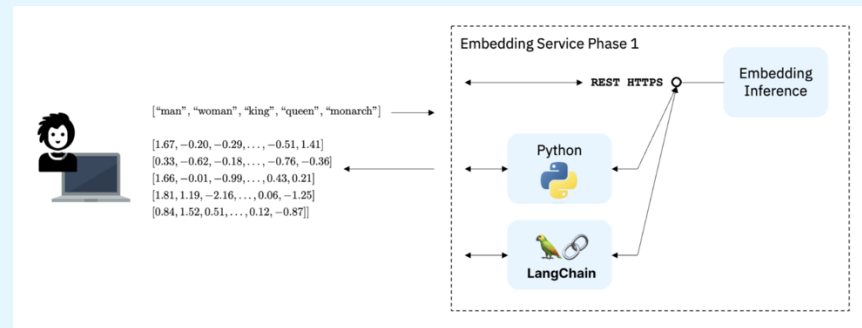
watsonx.ai Embeddings API

Convert text into dense vector embedding representations. Embeddings capture nuanced semantic and syntactic relationships between words and passages which are then stored in a vector database for retrieval.

Key benefits:

- Support RAG patterns with contextual grounding when utilizing embedding models for query and passage vectorization
- Improve RAG performance with semantically faithful representation of content, especially when compared to basic keyword-based alternatives in classic NLP modeling
- Efficient storage and compute profiles of embeddings make them easily infusible into generative AI applications

Learn more in [documentation](#). See list of [supported embedding models](#).



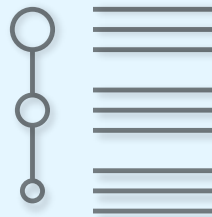
watsonx.ai Text Rerank API

Add precision to RAG use cases by improving the order of retrieved information with advanced re-ranking algorithms.

The Text Rerank API helps ensure relevant embeddings are used as context within a RAG pipeline, significantly enhancing performance.

- Reorder document passages (from most-to-least likely to answer) based on their similarity to a specified query
- Powered by the ms-marco-minilm-l-12-v2 reranker model ([see details](#))
- Fully integrated with the watsonx.ai REST API

Learn more in [documentation](#).

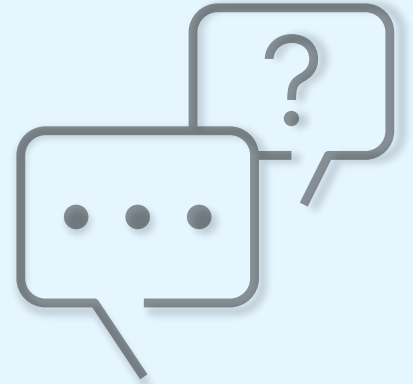


watsonx.ai Chat API

Create dynamic and adaptive conversational interfaces including chat-based application that require grounding documents, images, and tool calling for agent-driven applications.

- Build multi-user conversational workflows that use foundation models to generate answers.
- Easily identify different message types, such as a system prompt, user inputs, and foundation model outputs, including user-specific follow-up questions and answers.
- Supports granite, llama and mistral models (see [supported list](#))
- Fully integrated with the watsonx.ai API & SDK

Learn more about [adding chat functions](#) and [building agents](#) in documentation. See the [developer hub](#) for examples.



Agents

Three areas of agentic innovation

Pre-built agents

watsonx Orchestrate

Accelerate AI agent deployment.

Get started quickly with prebuilt AI agents powered with business logic and seamless integration to the tools that power your business.

Custom-built agents

watsonx.ai

Build custom-designed agents.

Design, deploy and manage AI agents with ease using pro- and low-code options.

Multi-agent orchestration

watsonx Orchestrate

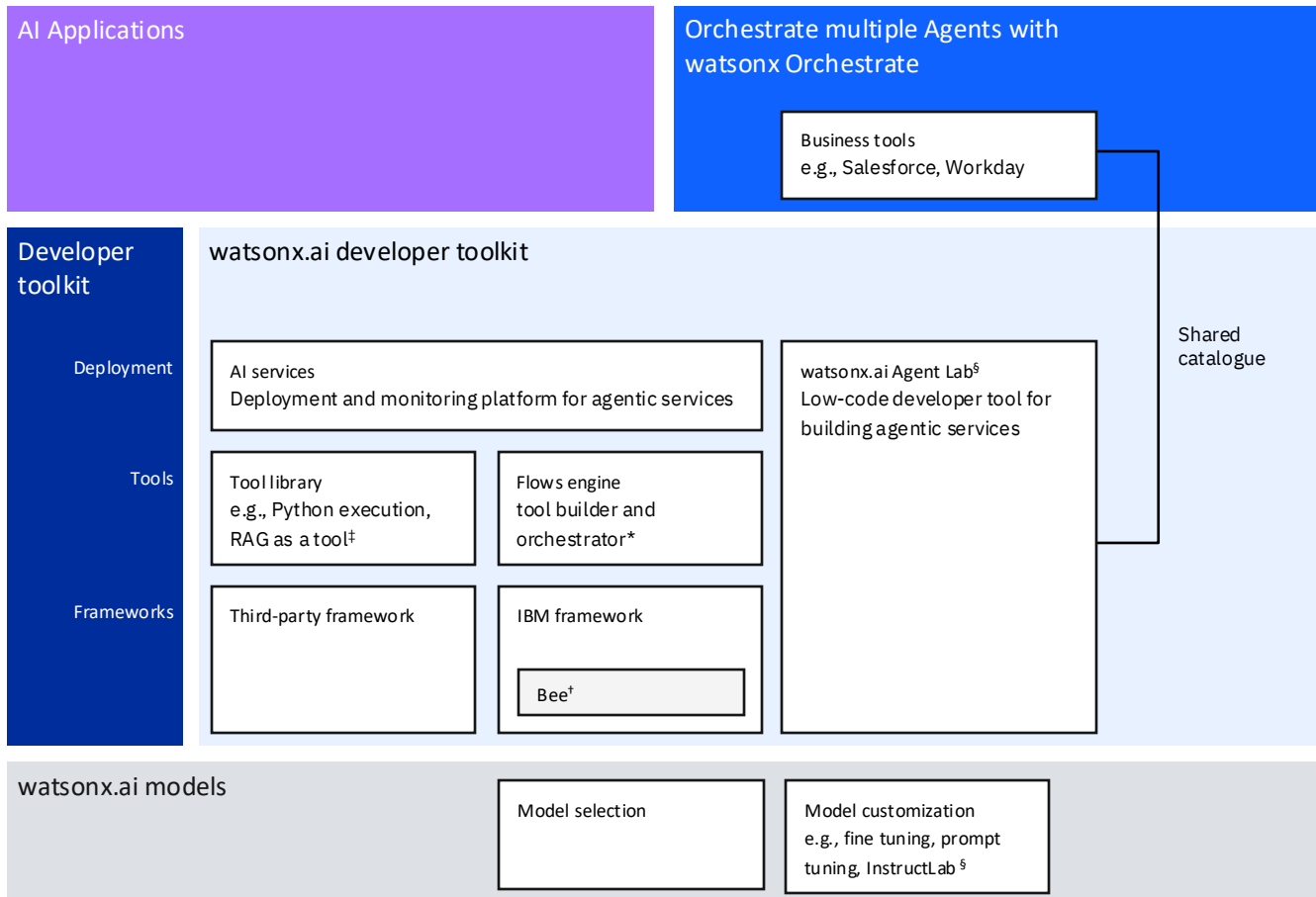
Manage all agents in one place.

Easily deploy and manage any agent for any task within a simple and unified user experience optimized to scale.

Architecture

Custom-built agents

One size does not fit all for building agents



* Stand-alone in tech preview; to be integrated into watsonx.ai

‡ Experimental, open-source project

§ In discovery

§ Coming soon

watsonx.ai Agents

What's available today?

- Support for Chat API, Tool Calling, JSON returns together with providing the foundation for agentic support
- AI Services that provide the foundation for deployment of python based agentic solutions
- End-to-end template for the development and deployment of a custom Langraph agents
- Introduction of Agent Lab (Beta), enabling a low-code experience to build agentic solution with 1-click deployment as a production-ready API endpoint

The screenshot displays the IBM watsonx AI Agents interface. The top navigation bar includes the 'IBM watsonx' logo, an 'Upgrade' button, a help icon, a notification bell, and user account information (IBM account, Dallas, and a profile icon labeled 'KB'). Below the navigation bar, the breadcrumb path shows 'Projects / Kate's sandbox /'. The main interface is divided into two panels: 'Build' on the left and 'Agent preview' on the right.

Build Panel:

- Set up:** A dropdown menu.
- Configuration:** A dropdown menu.
- Framework:** A dropdown menu with 'LangGraph' selected.
- Architecture:** A dropdown menu with 'ReAct' selected.
- Instruction:** A text area with a character count of '100/500'. The instruction reads: 'You are a helpful assistant that formats topics into structured GitHub issues. There are 4 types of issues, each with a specific template: 1. Epic: A large, broad objective that can be broken down into smaller actionable tasks. 2. Bug Report: A problem, malfunction or defect in the system.'
- Tools:** A section with two buttons: 'Add a tool +' and 'Create new tool'.
- Added tools (3):** A list of three tools:
 - Web search:** Retrieve information from the internet.
 - Document search: Vector index name:** Enables quick and efficient retrieval of information from vector indexes.
 - Python code executor:** Enables dynamic code execution.

Agent preview Panel:

- Github Agent | 12:34 pm:** A header for the preview.
- Welcome to Github Agent:** A message stating: 'To get started, paste in user paint points and I will help you turn it into a well formatted Github ticket.'
- Quick start samples:** Two buttons: 'Create a template for a new feature request with requirements.' and 'Create a template issue for a bug found in product.'
- Type something...:** A text input field at the bottom.

watsonx.ai SDKs

Build **agentic services** using popular open-source agentic frameworks through industry-standard API and SDK support.



Lightweight Engine

watsonx.ai Lightweight Engine

Reduced footprint, inference-only installation of watsonx.ai for client-managed software

Foundation Models

Access a variety of fit-for-purpose foundation models: IBM, open source, third party and imported custom foundation models.

Deploy Anywhere

Run watsonx.ai Lightweight Engine in any environment.

Reduced footprint

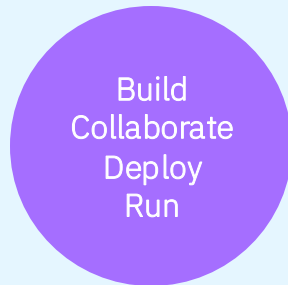
Run watsonx.ai Lightweight Engine with a smaller footprint.

Developer Experience

Leverage enterprise-grade APIs for a consistent developer experience across watsonx.ai full service and the lightweight engine.

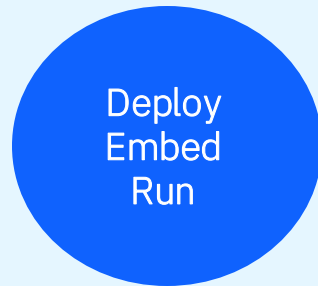
Software only

watsonx.ai



Full service, enterprise-grade AI toolkit for building and deploying end-to-end AI solutions.

watsonx.ai lightweight engine



lightweight engine to embed into external gen AI applications

watsonx.ai Lightweight Engine Features

Included

Features

- ✓ Foundation models (IBM, third-party, open-source)
- ✓ Imported custom foundation models
- ✓ watsonx.ai REST API
- ✓ watsonx.ai Text Embeddings API
- ✓ watsonx guardrails (HAP/PII filters)

Enterprise Capabilities

- | | |
|----------------------|-------------------------|
| ✓ Automated install | ✓ User Authentication |
| ✓ Horizontal Scaling | ✓ Logging |
| ✓ Backup and restore | ✓ Certification Manager |

Excluded, watsonx.ai full-service only

Features

- X Tuning Studio
- X Prompt Lab
- X Instruct Lab
- X Prompt Tuning via watsonx.ai REST API
- X Fine-tuning via watsonx.ai REST API ¹
- X Chat with Documents and Images

Enterprise Capabilities

- X Projects and Spaces
- X Access Control

¹ – on watsonx.ai full-service roadmap

Analyzing data and working with models

watsonx.ai Data Science and MLOps

Wide selection of integrated tooling to quickly build, deploy and manage machine learning models and AI applications

Build models

Comprehensive tooling for Model training and development

- Low/Pro code data science tools in a single environment
- Auto AI automates several aspects of the model lifecycle
- open-source libraries and integrated notebook-based interfaces
- Support for popular languages such as Python and R
- Integrated tooling for data pipelines and data preparation

Deploy models

Support for model deployment and management

- One-click deployment
- Continually deliver models into production
- Automated version control, retraining, code snippet generation for developers
- Build on OpenShift for reliability

Monitor models with watsonx.governance

Automate governance to drive trust across the AI lifecycle

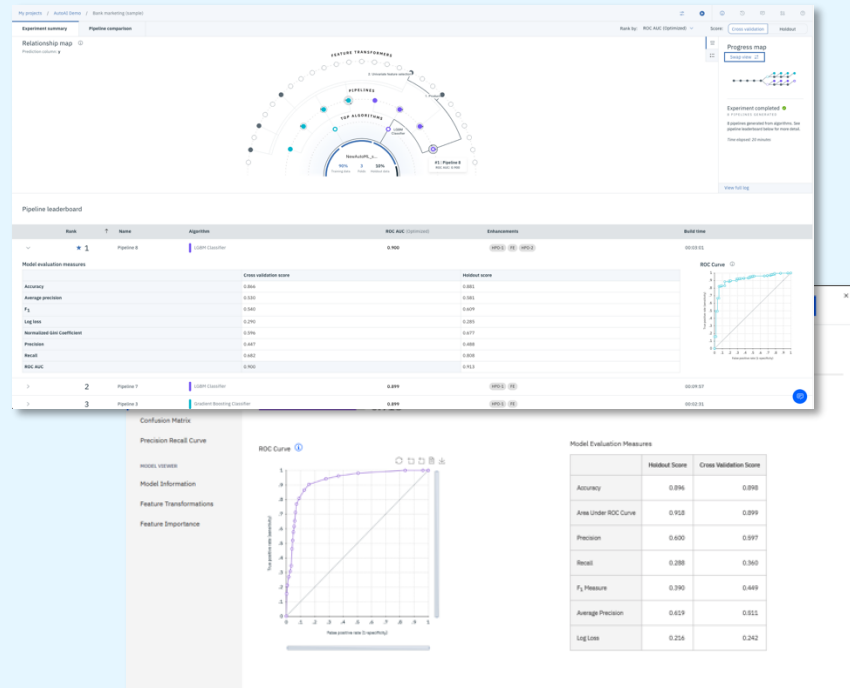
- Monitor drift over time, automatic retraining as required
- Provide model verification to stakeholders to ensure fairness, transparency and explainability over time
- Inform stakeholders of accuracy shifts before an issue occurs

watsonx.ai AutoAI

Expedite the creation of machine learning models with AutoAI, an easy to use, no-code interface. With a few clicks, quickly build powerful models such as classification, regression, time series forecasting, and more.

- Enhance the efficiency of the limited resources in a Data Science team by automating modeling.
- Quickly identify the best features, transforms, and models, as well as train the models, tune the hyperparameters, and rank the best-performing pipelines.
- Save an experiment as a notebook to understand the transformations applied to build the model or make changes.
- One click deployment of models, online or in batch mode

Learn more in [documentation](#)



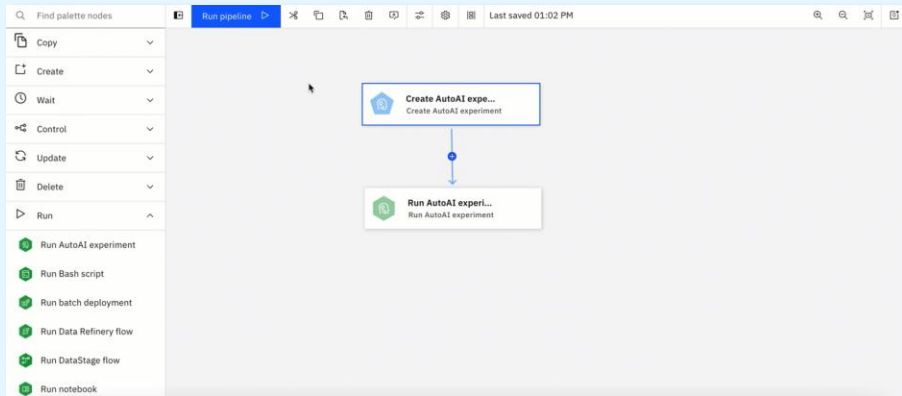
watsonx.ai Pipelines

Create end-to-end flow of assets from creation through deployment on a graphical canvas.

Key benefits:

- Increase efficiency and time savings with automated pipelines
- Combine tasks from different tools
 - Notebooks
 - Data refinery
 - Model creation
 - Model deployments
- Automate model training with new data sets
- Automatically deploy better challenger models

Learn more in [documentation](#)



watsonx.ai Synthetic Data Generator

Generate synthetic tabular data to address your data gaps

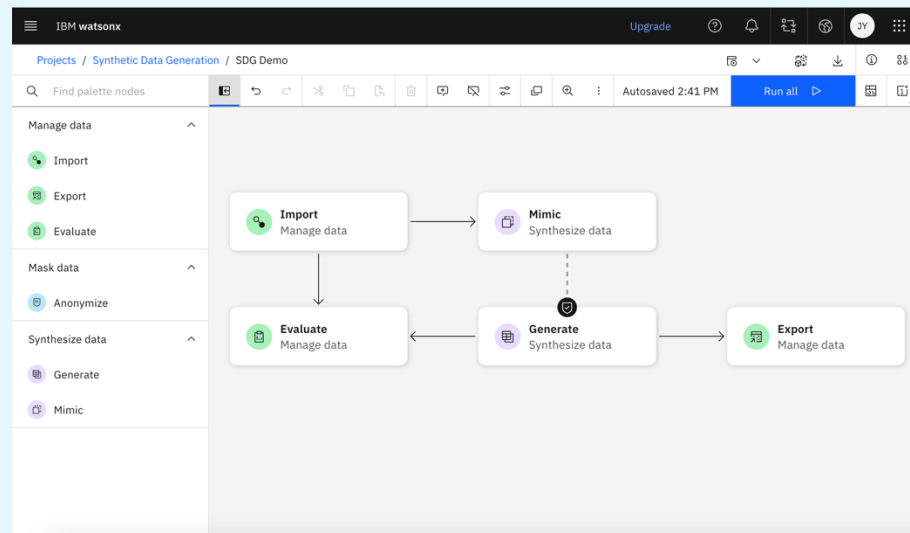
Create synthetic data at scale

- Unlock your valuable insights by using synthetic data.
- Create synthetic data using your existing data in a database or by uploading a file. Optionally, design your own data schema.
- Address data gaps and create synthetic edge cases to expedite classical AI model training.

Select your model & privacy needs

- Depending on your cost, fidelity, application, or data needs, you can select from multiple IBM models* to create your synthetic tabular data.
- When using existing data, IBM models apply differential privacy to minimize your privacy risk and give you control over the level of privacy protection required for your organization.

Learn more in [documentation](#)



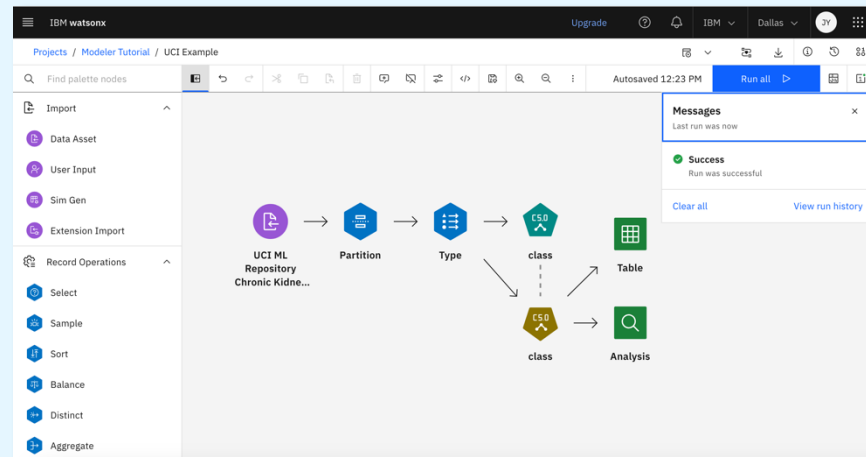
watsonx.ai Visual Modeling

The integrated SPSS Modeler brings a comprehensive set of nodes for data preparation, machine learning including auto modeling, data visualization, and push to production options.

Key benefits:

- Quickly build machine learning models using powerful drag and drop data analysis and visualization
- Drive direct insights from raw texts, uncover hidden value and patterns from text for prediction
- Deploy SPSS streams to into production

Learn more in [documentation](#)



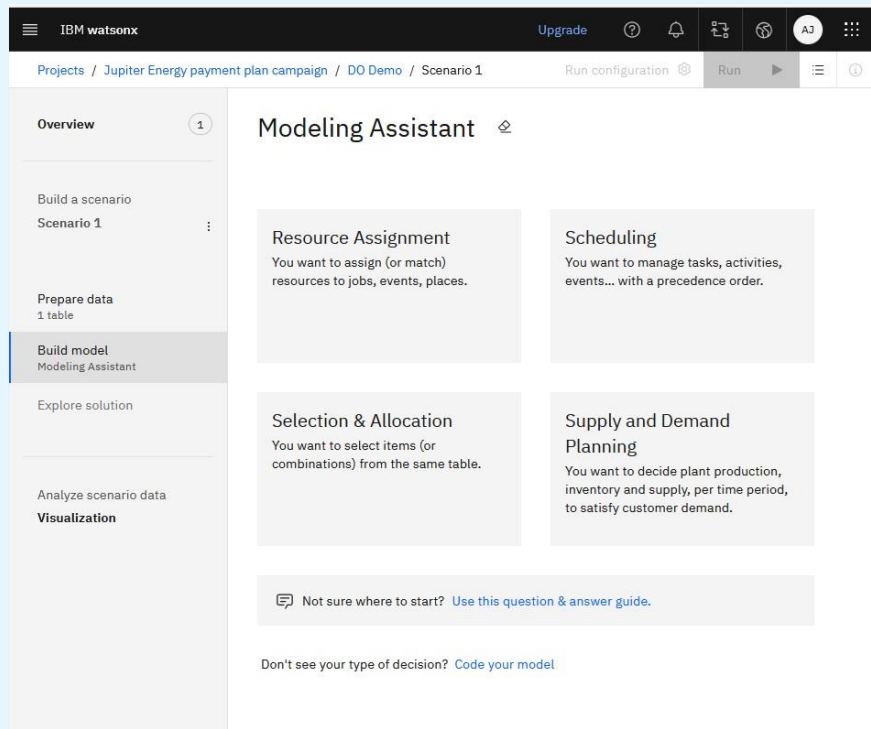
watsonx.ai Decision Optimization

Leverage the power of prescriptive analytics with Decision Optimization (DO). Evaluate a problem holistically, identify constraints, and determine the best plan to accomplish your goal (e.g. max profit, min costs, optimal scheduling, etc.)

Key benefits:

- Solve optimization models by using DO's highly powerful mathematical and constraint programming engines
- Improve productivity by validating optimization models more quickly and easily using visual dashboards
- Create innovative solutions by combining optimization technology with data science techniques such as machine learning
- Operationalize your projects by deploying optimization models in production to drive business impact around real use cases

Learn more in [documentation](#)



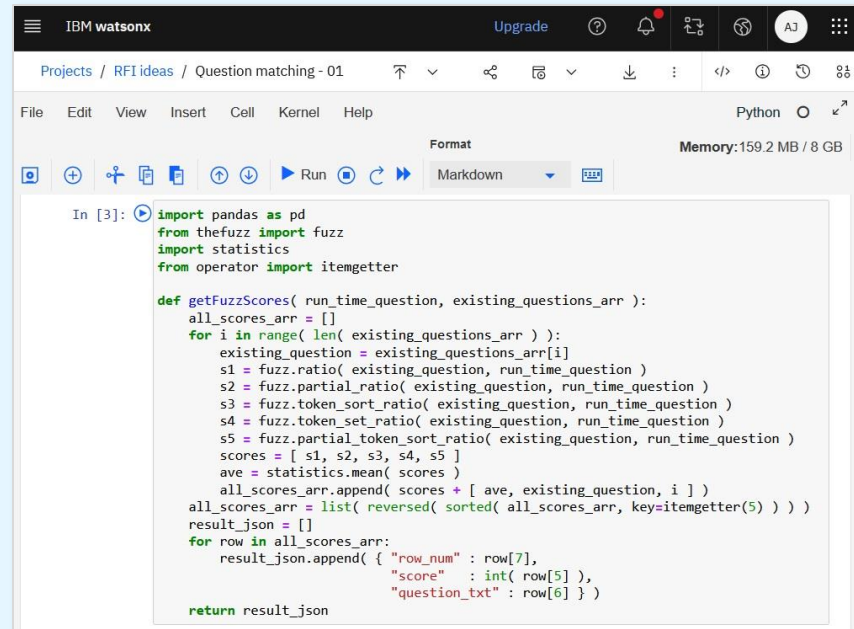
watsonx.ai Pro Code Tooling

Create, edit and execute Python and R code using Jupyter notebooks and scripts in code editors, for example the notebook editor or an the Rstudio integrated development environment (IDE)

Key benefits:

- Preinstalled IBM and opensource libraries
- Import your own libraries
- Connect notebooks to pipelines, publish to a catalog or Github for sharing and collaboration
- Automate repeatable processes through jobs to run the notebooks

Learn more in [documentation](#)



The screenshot displays the IBM watsonx Pro Code Tooling interface. At the top, there's a header bar with the 'IBM watsonx' logo, an 'Upgrade' button, and several utility icons. Below the header, a breadcrumb trail shows 'Projects / RFI Ideas / Question matching - 01'. The main interface features a menu bar with 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', and 'Help'. To the right of the menu bar, it indicates the current language is 'Python' and shows the memory usage as 'Memory: 159.2 MB / 8 GB'. Below the menu bar is a toolbar with various icons for file operations and code execution. The central area is a code editor displaying a Python script. The script starts with imports for 'pandas', 'fuzz', 'statistics', and 'operator'. It then defines a function 'getFuzzScores' which takes 'run_time_question' and 'existing_questions_arr' as arguments. The function iterates over 'existing_questions_arr', calculating various fuzz ratios (fuzz.ratio, fuzz.partial_ratio, fuzz.token_sort_ratio, fuzz.token_set_ratio, fuzz.partial_token_sort_ratio) and averaging them. It appends these scores to 'all_scores_arr'. Finally, it sorts 'all_scores_arr' and returns a 'result_json' object containing 'row_num', 'score', and 'question_txt' for each row.

```
In [3]: import pandas as pd
from thefuzz import fuzz
import statistics
from operator import itemgetter

def getFuzzScores( run_time_question, existing_questions_arr ):
    all_scores_arr = []
    for i in range( len( existing_questions_arr ) ):
        existing_question = existing_questions_arr[i]
        s1 = fuzz.ratio( existing_question, run_time_question )
        s2 = fuzz.partial_ratio( existing_question, run_time_question )
        s3 = fuzz.token_sort_ratio( existing_question, run_time_question )
        s4 = fuzz.token_set_ratio( existing_question, run_time_question )
        s5 = fuzz.partial_token_sort_ratio( existing_question, run_time_question )
        scores = [ s1, s2, s3, s4, s5 ]
        ave = statistics.mean( scores )
        all_scores_arr.append( scores + [ ave, existing_question, i ] )
    all_scores_arr = list( reversed( sorted( all_scores_arr, key=itemgetter(5) ) ) )
    result_json = []
    for row in all_scores_arr:
        result_json.append( { "row_num" : row[7],
                             "score" : int( row[5] ),
                             "question_txt" : row[6] } )

    return result_json
```

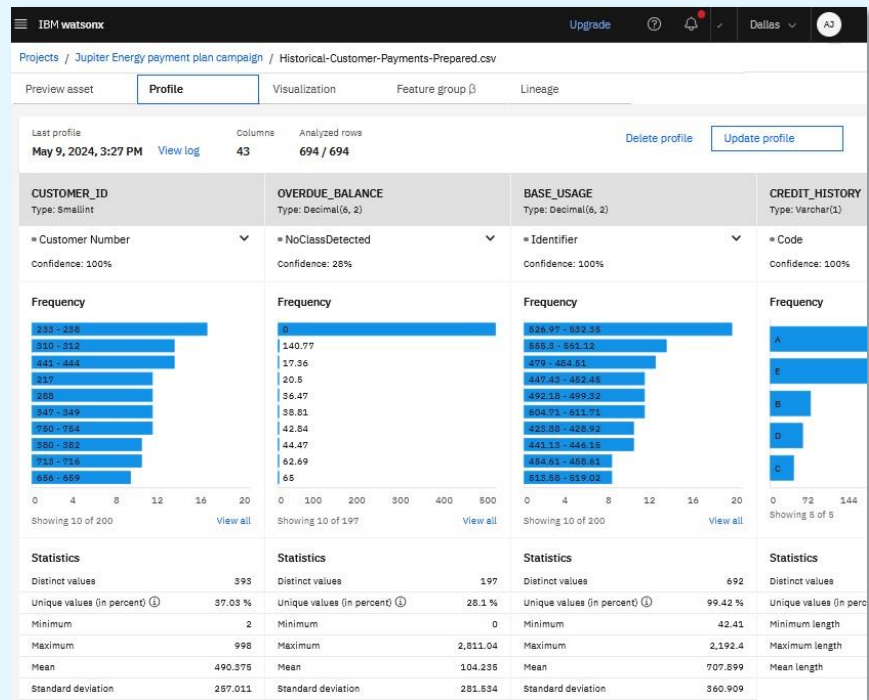

watsonx.ai Data Refinery

Conveniently analyze and prepare data for model training and all your AI application use cases within a single environment.

Key benefits:

- Retrieve or view tabular data from over 60 built-in data connectors
- Quickly analyze and visualize data with built-in data profiling and charting tools
- Cleanse and shape data with graphical editing with over 100 built-in operations
- Save time with scheduled flows and jobs for repeatable operations

Learn more in [documentation](#)

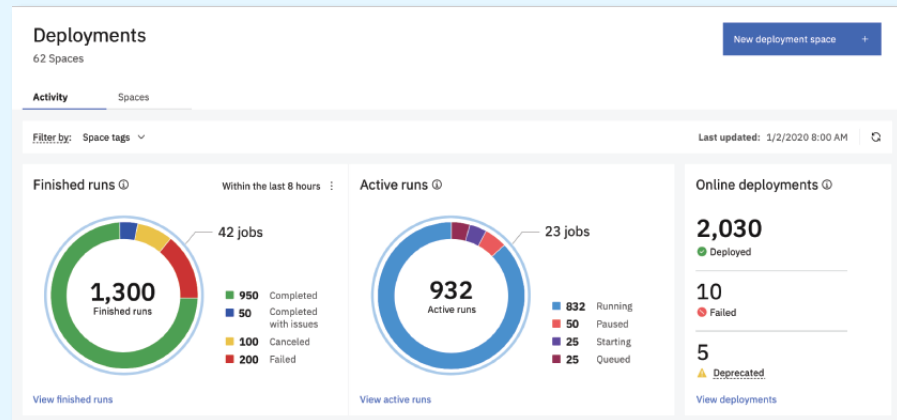


watsonx.ai Model Deployment

Expedite deployment of machine learning models, scripts, functions, and prompt templates for generative AI models.

- Secured with authentication credentials
- Support popular frameworks
- Deploy models in online or batch mode
- Score data through REST API calls
- Automate deployments with jobs

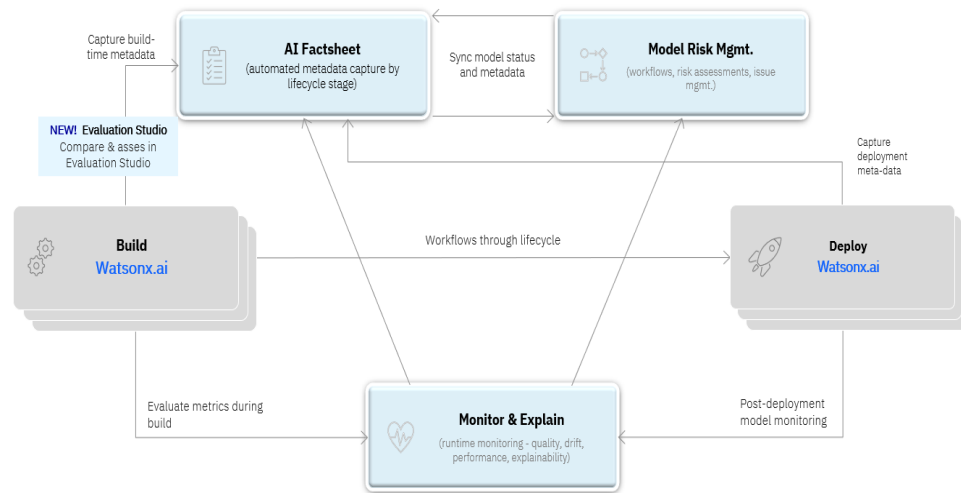
Learn more in [documentation](#)



watsonx.governance and watsonx.ai

Watsonx.governance brings **observability, explainability, and risk management** - key elements to build and deploy LLM applications and ML models.

- ✓ **Easily track and organize** AI Assets in Projects and Spaces by use case and associated lifecycle stage
- ✓ Enable AI Engineers to **easily determine the right Prompt asset** through experiments in Evaluation Studio
- ✓ **Evaluate quality and performance** of multiple task types (summarization, context generation, Q&A and more) during build with out-of-the-box metrics and integrated experience in Prompt Lab
- ✓ **Runtime monitoring** for quality, drift, and performance (and bias for ML models); user is alerted when thresholds are breached
- ✓ **RAG Root Cause Analysis insights** help AI engineers understand LLM responses
- ✓ **Assess hallucination** through quantifiable metrics for RAG task-type; **easily see** the phrases in **context used to generate responses**
- ✓ Reduce overhead of manual documentation through **automated documentation** of Prompt Template metadata and training input from the Prompt Lab

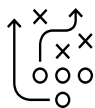


... and more!

- Out-of-the-box workflows through lifecycle
- Issue management integrated with use case
- Use case risk assessment
- Regulatory applicability assessments
- View AI use cases and deployments use across enterprise

Four ways to get started with watsonx.ai today

Looking forward to today's discussions



Free trial

Get started quickly by accessing AI models, data, and tools to build AI applications.



watsonx Developer Hub

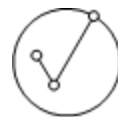
Access a library of templates and guides to get started building gen AI applications.



Request a client briefing or demo

Discussion and custom demonstration of IBM's generative AI watsonx point of view and capabilities. Understand where generative AI can be leveraged now for impact in your business.

2-4 hours onsite or virtual



5-year business value assessment

Engagement with an IBM multi-disciplinary team to jointly innovate and rapidly prove the business value of generative AI solutions using watsonx.

1-4 weeks

